

1. Difference Between in "Mathematics" and "Deep Learning":

- A. the variable  $x$  in Ch2-2 Note: Concept of Derivative in "Mathematics" and Ch2-3 Note: Gradient Descent Method in "Mathematics", need to be replaced to variable  $w$  (weight) in "Deep Learning"
- B. the function  $f$  in Ch2-2 Note: Concept of Derivative in "Mathematics" and Ch2-3 Note: Gradient Descent Method in "Mathematics", need to be replaced to function  $z$  (weighted sum function) in "Deep Learning"
- C. in "Deep Learning", input ( $x$ ) is constant not variable and weight ( $w$ ) is variable not constant